

Chinmay Narolia

+91-xxxxx-xxxxx | chinmay91797@gmail.com | github.com/cxinu | cxinu.github.io

ABOUT

Systems Software Engineer focused on systems programming, hardware-level interactions, and performance optimization. Experienced in building custom version control systems, GPU-accelerated pipelines, and high-performance distributed libraries. Proficient in C/C++, CUDA, Odin, and Go, with a strong interest in computer systems architecture, AI infrastructure, and OS internals.

EDUCATION

International Institute of Professional Studies, DAVV
Masters in Computer Application

Indore
2022 — 2027

PUBLICATIONS

- **A Lightweight Machine Learning Framework for DDoS Attack Detection in SDN**
Published in Springer | DOI: [10.1007/978-3-032-16791-0_36](https://doi.org/10.1007/978-3-032-16791-0_36)

PROJECTS

CUDA, C++, OpenCV, CudaCV – GPU-Accelerated Vision Pipeline 2026 — Present

- Engineered a GPU-accelerated image and video preprocessing pipeline using CUDA and OpenCV for real-time computer vision
- Implemented hardware-accelerated classic vision operators including adjustable Gaussian blur and Sobel edge detection
- Designed a modular architecture supporting real-time parameter tuning via keyboard controls and multi-mode processing streams

Odin, File I/O, Hashing (SHA-256), norse – A Git-like Version Control System 2025 — Present

- Implemented custom version control system replicating Git-style architecture using SHA-256 blob storage
- Designed internal file tracking system with `.norse/` metadata directory and staged object system
- Wrote foundational commands from scratch in Odin system programming language to master VCS internals and content-addressable storage

Lua, Redis, OpenResty, rllib – Distributed Rate Limiting Library 2025 — Present

- Built a high-performance, distributed rate-limiting library for OpenResty with Redis as backend
- Implemented fixed-window rate limiting with optional local caching via `lua_shared_dict` to reduce Redis load
- Published package to LuaRocks for community use

SKILLS

- **Programming Languages:** C/C++, CUDA, Odin, Go, Python, Lua, Shell
- **Systems & OS:** Linux Systems Programming, OS Internals, GPU Programming, TCP/IP stack, Content-Addressable Storage
- **Infrastructure & Tools:** OpenCV, PyTorch, MLIR, Docker, K8s, Redis, Neovim, Odin Toolchain

CERTIFICATIONS & ONGOING LEARNING

- **Nand2Tetris:** Coursera Certified ([Verify](#)) — Constructed a basic computer system from fundamental logic gates up to an operating system. (Jan 2026)
- **Advanced Architecture:** Deepening OS and hardware-software interface knowledge via **Computer Systems: A Programmer's Perspective (CS:APP)**.
- **GPU Programming:** Exploring low-level AI infrastructure tooling, including CUDA, MLIR, and PyTorch internals.

ACHIEVEMENTS

- **GATE Qualified (2026):** Successfully cleared the national-level Graduate Aptitude Test in Engineering.
- **HackTheBox CTF (2025):** Ranked 458th out of 8,130 teams globally with Team Exploit Academy.